

Aditya Kulkarni

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Raleigh, NC (open to relocation to SF)

PROFESSIONAL SUMMARY

Backend engineer with production AI/ML experience in TypeScript and Java. At Isomer AI, shipped 10+ backend features and 7 production deployments -- multi-model LLM routing across Claude, GPT-4o, and Gemini with MCP tooling, Temporal.io workflow orchestration, and microservices observability via Sentry; 394 commits in a 3-person team. Built a concurrent inventory reservation API in Java 21 + Spring Boot with Testcontainers-verified transactional correctness. MS CS at NC State (May 2026). IEEE-published researcher. HackNC24 Top 10. Case studies: kulkarniaditya.com

CORE COMPETENCIES

TypeScript / Node.js

Java 21 / Spring Boot

React

PostgreSQL / Redis

Microservices

REST API Design

Temporal.io

Kafka

LLM Pipelines (Claude, GPT-4o, Gemini)

MCP Tooling

Observability (Prometheus, Sentry, OTEL)

AWS / GCP

Docker / CI/CD

Distributed Systems

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Shipped 10+ backend features across 7 production deployments for an AI-powered claims automation platform; 394 commits across TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, and Pusher in a microservices architecture with observability via Sentry
- Built multi-model AI agent tooling routing across 3 LLM providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models using MCP; implemented Temporal.io workflow orchestration, RBAC, real-time event pipelines, document processing (PDF, email via Microsoft Graph API, S3), and Auth0 authentication

NC State University April 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform with JBrowse 2 serving 500GB of strawberry and grape datasets to 100+ researchers worldwide on Google Cloud Run
- Applied hyperparameter tuning and time-series analysis to ML architecture, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%
- Automated deployments on AWS and GCP; integrated monitoring dashboards with Grafana; supported model red-teaming for reliability

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a data-driven system automating support ticket analysis, reducing processing time by 15%
- Built distributed data pipelines with Hadoop + Spark for batch processing and Kafka streaming for real-time ingestion, cutting ticket-processing latency by 20%

PROJECTS

[Inventory Reservation Service](#) Java 21 / Spring Boot

Transactional inventory reservation API preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations. Idempotent POST endpoints with RFC 9457 error responses; automated expiry via scheduled cleanup. Verified concurrency correctness with Testcontainers: two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Security, Testcontainers, Docker

[PulseRank - Real-Time Leaderboard & Activity Feed](#) Distributed Systems

Event-driven leaderboard and activity feed: MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed document projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads; live score updates over Server-Sent Events. Multi-store consistency validated with Testcontainers; hot paths benchmarked with k6; OpenTelemetry + Prometheus observability.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

[LLM Eval Harness](#) AI/ML

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types on a 2,000-example frozen test set with 3 repeated runs. Tracked ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call per model. Statistical significance via paired bootstrap CIs, McNemar test, and Holm correction. JSONL + Parquet storage; DuckDB analytics; Streamlit dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

[AlgoPulse - Agentic AI Platform](#) Event-Driven / Scale

Identified API-polling bottleneck under load; pivoted to event-driven streaming with Supabase Edge Functions. Temporal.io for durable workflow state; Kafka for worker coordination and fault tolerance. Reached 2,000 async inference requests/second without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

[VectorLift - Semantic Search Engine](#) Search / ML

Production-grade semantic search on MS MARCO: BM25 baseline, dense bi-encoder retrieval (sentence-transformers fine-tuned with in-batch negatives), and cross-encoder re-ranker. Kafka-driven indexing pipeline into Elasticsearch and Qdrant; FastAPI serving with per-stage latency tracking; NDCG@10/MRR@10/MAP evaluation with bootstrap significance testing.

Python, PyTorch, Elasticsearch, FAISS, Qdrant, Kafka, FastAPI, Streamlit, Prometheus, Grafana

[Mapify - ML & NLP Tool](#) IEEE Published

T5-transformer pipeline converting 1,500+ PDFs and images into mind maps and flowcharts. Reduced academic review time by 60%; NLTK and NetworkX for structure extraction. Presented to 500+ attendees at IEEE conference.

Python, T5-transformer, NLTK, NetworkX

EDUCATION

[North Carolina State University -- MS Computer Science](#) Aug 2024 - May 2026

Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms

[Mumbai University -- BTech Computer Science](#) Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain

AWARDS & LEADERSHIP

[HackNC24 Top 10](#) -- GenAI Finance App (PerCent); ranked top 10 of 100+ teams in 48-hour build 2024

[Sanjeevani Health App](#) -- Top 1% of 1,100+ submissions (top 12); 87% ML classification accuracy 2023

[IEEE Publication](#) -- Mapify: T5-transformer pipeline for PDF-to-mind-map conversion; 500+ attendees 2022

[Google Developer Student Club Lead](#) -- Led 4+ tech events (hackathons, conferences) reaching 500+ students 2022-2024

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, JavaScript, SQL, C, R **Backend:** Node.js, Spring Boot, Express, NestJS, FastAPI, REST APIs

Frontend: React, Next.js, Tailwind CSS **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant

AI/ML: LangChain, MCP, Hugging Face, PyTorch, TensorFlow, RAG, OpenAI/Anthropic/Google SDKs

Infrastructure: AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Spark, Hadoop **Observability:** Prometheus, Grafana, Sentry, OpenTelemetry

DevOps: GitHub Actions, Flyway, Testcontainers, Docker Compose